NGUYEN CAO QUOC

MIDDLE AI ENGINEER

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Summary

Passion in crafting useful AI products with high performance and accuracy. Have 3 years in developing native AI applications, skilled in many modern AI technologies, experienced with concurrency programming, performance optimization. Also have experiences in researching and evaluating AI solutions. Currently pursuing a Master's degree in Computer Science for wider and deeper AI knowledge.

Skills

Core AI Concepts	Deep Learning (Computer Vision, NLP), Machine Learning, Generative AI, RAG, AI Agents, Model Context Protocol (MCP)
Languages	Python (willing to work with JavaScript/TypeScript)
Frameworks & Libs	PyTorch, TensorFlow, LangChain, LangGraph, LlamaIndex, FastAPI, FastMCP, PydanticAI, OpenAI SDK
Cloud & Tools	Azure (Foundry, OpenAI, Document Intelligence, Blob Storage), AWS (EC2, S3, Bedrock), Docker, Git, MCP, Agent-to-Agent (A2A)
Databases	PostgreSQL, MongoDB, Milvus, FAISS, Weaviate

Professional Experience

Saigon Technology *Middle AI Engineer* Mar 2026 – Present

Content Moderation & Document Intelligence Platform

- Built a real-time AI-powered content moderation and document intelligence backend, handling multi-modal content safety (text, image, PDF), automated document parsing with badge assignment, and an AI posting assistant through both synchronous and asynchronous REST APIs.
- Designed and implemented an **N-Layer architecture**, refactoring over **2000+ lines** of monolithic code into focused, testable and maintainable modules.
- Reduced content moderation processing time by approximately **83%** by converting sequential moderation tasks into parallel execution using **asyncio** concurrent pipelines and batch processing.
- Built a multi-modal **content moderation pipeline** supporting **text, image and PDF**, with each modality implemented as an independent **Strategy** class.
- Implemented text moderation by combining traditional algorithm-based detection with **Azure AI Content Safety** to detect sexual content, violence, hate speech and self-harm.
- Developed an image moderation pipeline integrating **Azure AI Content Safety**, **OpenAI Vision** for QR code detection, and **Azure Document Intelligence OCR** for embedded text extraction and re-analysis.
- Developed an AI-powered document parsing service using **Azure Document Intelligence OCR** and **OpenAI** for automated badge assignment based on extracted qualifications, including confidence scoring and manual review workflow.
- Containerized the application with **Docker**, ensuring production-grade deployment using **Uvicorn** with a single application entrypoint.
- Leveraged **Azure Cloud, OpenAI, Azure AI Content Safety, Azure Document Intelligence, Azure Foundry, Blob Storage, FastAPI, Python** and **OpenCV**.

TMA Solutions *AI Engineer* Jan 2025 – Mar 2026

Agentic Platform

- Implemented **AI agents for healthcare** applications to deliver intelligent solutions that improved adaptability to industry needs.
- Took responsibility for developing a **dynamic language module** that enabled adaptive and context-aware for agent behavior.
- Engineered intelligent workflows on **ophthalmology** to enhance process automation and improve clinical decision support.

Clinical Contracts Extraction and Invoice Matching

- Took responsibility for building the foundational architecture of the AI services.
- Leverage **Azure Cloud**, **LangGraph** and **Pydantic Agent** to create Contract Extraction Agent and Invoice Validation Agent.
- Ensuring the output format of Agents for embedding into **Milvus** database.
- Ensuring correctness and completeness of information before the invoice-matching step.

- Implemented concurrent AI Agent service flows, improve execution speed by **50%**.
- Designed and refined prompt instructions, and used **Pydantic** to enforce strict JSON output formats from LLMs.
- Leveraged **TOON** format in specific cases to reduce token usage.
- Collaborated with the tech lead to design the invoice–contract matching logic, improving overall correctness accuracy by **40%**.

VINAI RESEARCH *AI Engineer Intern* Sep 2024 – Jan 2025

Face Recognition System

- Researched and integrated **ArcFace Loss** into Deep Learning models to improve **Face Identification** performance.
- Evaluated models using **TAR@FAR** metrics, accuracy on **IJBB/IJBC** and compare against baselines.
- Analyzed **Cosine Scheduler**'s impact on learning rate adjustment and model convergence.
- Used **FastAPI** to develop interactive demos to show my solution's capabilities.

VIETVANG CONSULTING *AI Engineer* Jan 2024 – Sep 2024

Stock Keeping Unit Detection and Counting System

- Created and labeled image datasets using **Labelme**, ensured reliable training data for SKU detection and counting.
- Researched **YOLOv10** architecture, compiled detailed documentation to provide technical reference for project team.
- **Fine-tuned model** on custom datasets and evaluated performance. Increased model accuracy on real-world data.
- Developed interactive **POC** to showcase **YOLOv10** capabilities, helping customers understand AI applications and contexts.
- Used **FastAPI** to implemented APIs to support development teams in building customer-facing application.

Education

2024 – Present **Master of Computer Science**
University of Information Technology, VNU-HCM

2020 – 2024 **Bachelor of Computer Science**
University of Information Technology, VNU-HCM

Achievements

- IELTS 6.0 (English Certification) - View Certificate
- IBM AI Engineer (Coursera Professional Certificate) - View Certificate
- IBM Data Scientist (Coursera Professional Certificate) - View Certificate

Publications

Explainable Intelligence in Digital Twins (EIDT - Scopus-indexed conference)
Le Xuan Tung and **Nguyen Cao Quoc**: *Leveraging Large Language Models as Faithful Explainer for Text Classification* - EIDT 2025.
Link: Springer Lecture Notes in Electrical Engineering

Honors and Awards

FIRST PRIZE OF SOICT HACKATHON 2024 **AI POWERED APPLICATION** Nov 2024 – Jan 2025
Chatbot for processing administrative documents

- Utilized **Python** and **Llama-Cloud**'s API to process raw data into the required format, ensuring data consistency and quality.
- Leveraged **Llama-Index**'s API to convert data into markdown format and renamed files according to specified naming conventions.
- Employed **OpenAI**'s API to develop an API capable of generating **FAQ** pairs related to file content and file names.
- Conducted quality checks and evaluations of the generated FAQ pairs, **fine-tuning** them to ensure accuracy and relevance.